

Algebra general exam. January 13th 2012, 9am-2pm

Directions.

- Please show all your work and justify any statements that you make.
- State clearly and fully any theorem you use.
- Vague statements and hand-waving arguments will not be appreciated.
- You may assume the statement in an earlier part proven in order to do a later part.

DO EACH PROBLEM ON A SEPARATE SHEET OF PAPER, AND STAPLE THEM TOGETHER IN THE CORRECT ORDER BEFORE TURNING THE EXAM IN.

Problem 1

1. Let $F = \mathbb{F}_q$ be a finite field, where $q = p^r$ is a power of a prime p . Let $G = GL_n(F)$ be the group of all $n \times n$ invertible matrices with entries in F . Once you pick an ordered basis of $V := F^n$, you may find it useful to identify G with the group of invertible linear operators on V .
 - (a) (6 pts) Calculate the order of G . Explain your answer carefully and write it in the simplest form as you can.
 - (b) (3 pts) Determine the order of a Sylow p -subgroup of G , and explicitly exhibit a Sylow p -subgroup U of G .
 - (c) (2 pts) What is the normalizer in G of the Sylow p -subgroup U that you exhibited in (b)? An answer is sufficient.
 - (d) (3 pts) How many Sylow p -subgroups of G are there? Explain how your answer in (d) is consistent with Sylow's theorem.

Problem 2

2. Let G be a subgroup of the symmetric group S_n for some integer $n > 1$. Assume that G acts transitively on $\mathbf{n} := \{1, 2, \dots, n\}$, that is, for any $i, j \in \mathbf{n}$ there exists $g \in G$ s.t. $g(i) = j$.

A partition of $\mathbf{n}$ is a decomposition $\mathbf{n} = X_1 \cup \dots \cup X_m$ into a disjoint union of nonempty subsets. There are two trivial partitions: $\mathbf{n} = \mathbf{n}$ and $\mathbf{n} = X_1 \cup \dots \cup X_n$ (so each X_i has just one element). Otherwise the partition is said to be nontrivial. The group G is called imprimitive if there is a nontrivial partition $\mathbf{n} = X_1 \cup \dots \cup X_m$ such that, for $g \in G$ and $1 \leq i \leq m$, $g(X_i) = X_j$ for some j . (That is, G permutes the partition members among themselves.) The set $\{X_i\}$ is called a system of imprimitivity for

the action of G on $\mathbf{n}$. The group G is called primitive if it is not imprimitive.

- (a) (3 pts) Let $n = 6$ and consider the cyclic subgroup $G := \langle (1, 2, 3, 4, 5, 6) \rangle$ of S_6 . There are two non-trivial systems of imprimitivity for the action of G on $\mathbf{n}$. Find them.
- (b) (3 pts) Prove that if $X_1 \cup \dots \cup X_m$ is a system of imprimitivity for the action of G on $\mathbf{n}$, then all subsets X_i have the same size n/m .
- (c) (4 pts) G is said to be doubly transitive if given elements $a, b, c, d \in \mathbf{n}$, with $a \neq b$ and $c \neq d$, there exists $g \in G$ such that $g(a) = c$ and $g(b) = d$. Show that a doubly transitive group G is primitive.
- (d) (4 pts) Show that if $n \geq 3$, the alternating subgroup $G = A_n$ of S_n is primitive.

Problem 3

Let $R = \mathbb{Z}[\sqrt{-2}]$.

- (a) (7 pts) Prove that R is a Euclidean domain. Hint: Use the square of the usual complex norm.
- (b) (8 pts) Write 7 and 11 as products of irreducible elements of R . Justify your answer.

Problem 4

Let R be a ring with 1 . The opposite ring R^{op} is defined as follows: as a set $R^{op} = R$, the addition on R^{op} coincides with the addition on R and the multiplication $*$ on R^{op} is the multiplication on R in reverse order, that is, $a*b = ba$ (where ba is the product in R). Let $e \in R$ be an idempotent element, that is, $e^2 = e$.

- (a) (2 pts) Prove that $eRe = \{ ere : r \in R \}$ is a subring of R .
- (b) (6 pts) Consider the left R -module $M = Re$. Prove that its endomorphism ring $End_R(M) = Hom_R(M, M)$ is isomorphic to $(eRe)^{op}$, the opposite ring of eRe .

Problem 5

(9 pts) Let F be a field, n a positive integer and $M_n(F)$ the set of $n \times n$ matrices over F . Let $A \in Mat_n(F)$ be such that $A^2 = A$. Prove that A is diagonalizable and classify all such A up to similarity. (Recall that $A, B \in Mat_n(F)$ are similar if there exists $C \in GL_n(F)$ s.t. $C^{-1}AC = B$.)

Problem 6

Let R be a commutative ring with 1 . Recall that a left R -module M is called Noetherian if it satisfies the ascending chain condition on submodules and Artinian if it satisfies the descending chain condition on submodules. Assume that an R -module M is both Artinian and Noetherian. (For example, R might be a field, and M might be a finite-dimensional vector space over R). Let $T : M \rightarrow M$ be an R -module

homomorphism.

- (a) (3 pts) Prove that there exists $k \in \mathbb{N}$ s.t. $\text{Ker}(T^k) = \text{Ker}(T^{2k})$ and $\text{Im}(T^k) = \text{Im}(T^{2k})$.
- (b) (4 pts) Prove that if k is as in part (a), then $M = \text{Ker}(T^k) \oplus \text{Im}(T^k)$
- (c) (2 pts) Deduce from (a) and (b) that there exist submodules M_0 and M_1 of M s.t. $M = M_0 \oplus M_1$, $T|_{M_0}$ is nilpotent and $T|_{M_1}$ is invertible (as a map from M_1 to M_1).
- (d) (5 pts) Now assume that R is a field of characteristic zero, M is a finite-dimensional vector space over R and $\text{tr}(T^n) = 0$ for every $n \in \mathbb{Z}_{>0}$. Prove that T is nilpotent. Hint: Apply (c), assume that $M_1 \neq 0$ and reach a contradiction by applying the Cayley-Hamilton theorem to $T|_{M_1}$.

Problem 7

If q is a prime power, denote by $\mathbb{F}_q$ a finite field of order q .

- (a) (6 pts) Find a monic irreducible polynomial of degree 3 over $\mathbb{F}_5$ and use it to construct a field of order 125 . Justify your answer.
- (b) (6 pts) Find all q for which the polynomial $p(x) = x^2 + x + 1$ is irreducible in $\mathbb{F}_q[x]$. Hint: What can you say about roots of $p(x)$ and what do you know about the multiplicative group $\mathbb{F}_q^\times$?

Problem 8

Let F be a field of characteristic zero, let K and L be finite extensions of F and KL the compositum of K and L .

- (a) (4 pts) Prove that $[KL : F] \leq [K : F] \cdot [L : F]$.
- (b) (2 pts) Assume that $[K : F]$ and $[L : F]$ are relatively prime. Prove that $[KL : F] = [K : F][L : F]$.
- (c) (4 pts) Give an example where $K \cap L = F$ but $[KL : F] \neq [K : F][L : F]$.
- (d) (4 pts) Assume that K/F and L/F are both Galois. Prove that $\text{Gal}(KL/F)$ is isomorphic to a subgroup of $\text{Gal}(K/F) \times \text{Gal}(L/F)$. (You need not prove that KL/F is Galois).

Note: The assertions of (a),(b) and (d) remain valid for F of positive characteristic, but part (a) has shorter proof in the case of characteristic zero.